

	DEPARTMENT OF MECHATRONICS AND INDUSTRIAL ENGINEERING			(A)	
	ODD semester TEST (1) Academic year 2023/2024				
	Subject	INTRODUCTION TO ENGINEERING SCIENCE			
	Class	B2MI/E & B4MI/E	Date	27 / 12 / 2023	
	Max. Marks	40	Time	1 hours	
Student Name		Class		Reg. No	

Question No.	Q1	Q2	Q3	Total
Maximum Marks	20	20	20	TOTAL Out of 40
Marks Scored				
Lecturer	Dr. Fawaz Ahmed Ghaleb		Signature	

NOTES: Answer **Question No. (1)** and any other **ONE** questions:

QUESTION (1) (20 Marks)

Fill the following statement

- is a broad discipline that encompasses many different scientific principles and associated mathematics that underlie engineering.
- Engineering Science aims to provide a comprehensive grounding in relevant to engineers by an ability to identify, formulate, and solve engineering problems by applying principles of engineering,, and mathematics.
- Engineers consider important factors such as and when designing these products.
- Good engineers have time-management skills that enable them to and
- Mechatronics & and industrial engineering developed in direct response to industrial demand for engineers with is a combination of control, and systems design engineering streams.
- Engineers are problem solvers. Once they have obtained a solution to a problem, they need to effectively their solution to various inside or outside their organization.
- The title of a report should be informative description of the
- Engineers play a significant role in the design and maintenance of a nation's infrastructure, including systems,, and transportation.

QUESTION (2) (20 Marks)

Match part I and II

I								II		
1-	It is a combination of mechanical, electronics, control, computer and systems design engineering streams.							A	Civil Engineering	
2-	It is the body of knowledge that explores the physical and natural world.							B	Engineering Science aims	
3-	It is another way of conveying information in a brief way to interested individuals.							C	Executive summaries	
4-	To provide the reader with copies of all original data sheets, diagrams, and supplementary notes							D	Electrical Engineering	
5-	An ability to communicate effectively with a range of audiences							E	Abstract	
6-	It is a broad discipline that encompasses many different scientific principles and associated mathematics that underlie engineering.							F	Engineering Science	
7-	It is a very important part of a report							G	Appendix	
8-	Are means of communicating to people in top management positions, such as a vice president of a company, the findings of a detailed study or a proposal.							H	Short memos	
9-	It is including structural, environmental, transportation, water resources, and geotechnical.							I	Mechatronics & Industrial Engineering	
10-	It is branches including power generation, power transmission and distribution, and controls.							J	Science	
I	1	2	3	4	5	6	7	8	9	10
II										

QUESTION (3) (20 Marks)

Explain briefly why?

1. Why is mechatronics popular?
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2. Why is basic science important in engineering?
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3. Why is an abstract important to the readers?
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4. Why are good oral and written communication skills important in engineers.
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5. Why is technical report important in engineering?
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